



$$\partial[v_0v_1v_2] = [v_1v_2] - [v_0v_2] + [v_0v_1]$$

$$\partial\partial[v_0v_1v_2] = \partial[v_1v_2] - \partial[v_0v_2] + \partial[v_0v_1]$$

$$([v_2] - [v_1]) - ([v_2] - [v_0]) + ([v_1] - [v_0])$$

$$= 0$$

every term cancels — a boundary has no boundary